

International Journal of Chemistry and Technology

<http://dergipark.org.tr/ijct>

Research Article



Virtual Screening and Docking of Natural Compounds as Selective Inhibitors of Mutant NRAS in Melanoma.

Nayanika DAS¹, Vijay BALADHYE^{1*}¹ Bioinformatics Centre, Savitribai Phule Pune University, Pune 411007 (MS), India.*Corresponding author : vijaybaladhye@gmail.com

ABSTRACT

Melanoma is an aggressive and currently incurable skin cancer, primarily affecting individuals with light skin pigmentation. Key oncogenic drivers include mutations in BRAF, NRAS, CDK4 and MITF, with signaling mediated mainly through the MAPK and PI3K–Akt pathways. MEK inhibitors have shown limited clinical success, emphasizing the need for novel therapeutic agents. In this study, structure-based virtual screening and molecular docking were employed to identify natural products that inhibit mutant NRAS. The crystal structure of wild-type NRAS was obtained from the Protein Data Bank and clinically relevant mutations (Gly12, Gly13, Gln61) were introduced computationally. A library of natural compounds from the ZINC database was screened, identifying candidates based on binding free energy. Eleven top-ranking compounds were further analyzed for binding interactions, revealing stabilizing hydrogen bonds and hydrophobic contacts, particularly involving residues Phe28 and Asp119. These compounds exhibited strong predicted affinity for the NRAS mutant binding site, representing promising scaffolds for further development. The results demonstrate the utility of computational approaches to accelerate early-stage drug discovery targeting oncogenic NRAS in melanoma.

Keywords: NRAS, melanoma, molecular docking, MEK inhibitor, natural compounds etc.

1. INTRODUCTION

Melanoma is a deadly form of skin cancer responsible for most deaths worldwide. At the basal level of the epidermis melanocytes are found, which produce the UV absorbing pigment melanin (Cancer.Net Editorial Board, 2025). It's a combination of environmental and genetic factors; the majority of melanomas arise from somatic mutations acquired later in life. Among the key molecular drivers of melanoma are mutations in the RAS family of small GTPases (Al Mahi & Ablain, 2022). RAS family proteins play a central role in a variety of cellular processes such as signaling, survival, apoptosis, and membrane trafficking (Cherfils & Zeghouf, 2013; Hu & Martí, 2025; Ostrem & Shokat, 2016). They function like binary switches, cycling between an active (GTP-bound) and inactive (GDP-bound) state, this switching mechanism is governed primarily by two flexible regions known as Switch I (residues ~30–38) and Switch II (residues ~59–67). (Hu & Martí, 2025; Kolch et al., 2023; Li & Zhang, 2004; Spoerner et al., 2010; Vetter &

Wittinghofer, 2001) and inactive (GDP-bound) states. Ras is activated or inactivated when bound to GTP or GDP, respectively. When activated, they can interact with effector proteins that control fundamental biological processes such as EGFR, MAPK, or PI3K (Ardito et al., 2012; Howe et al., 1992; Navas et al., 2012; Normanno et al., 2009; Rodriguez-Viciana et al., 1994; Wood et al., 1992). However, excessive activation of RAS will eventually lead to cancers such as melanoma, colorectal, lung, and especially pancreatic cancer (Li et al., 2018; Moore et al., 2020). Mutations of RAS key residues such as G12, G13 and G61, are found in more than 30% of human tumors. Most oncogenic mutations in Ras occur in the residues G12, G13 and Q61, which are located in a region conserved between HRAS, KRAS and NRAS (Pylyayeva-Gupta et al., 2011). Among the three RAS isoforms (HRAS, KRAS, and NRAS), NRAS plays a prominent role in melanoma (Johnson & Puzanov, 2015; Wellbrock & Arozarena, 2016). Activating mutations in NRAS, particularly at codon 61, are observed in a substantial fraction of melanoma cases and are strongly

associated with aberrant activation of the MAPK signaling pathway (Johnson & Puzanov, 2015; Pylyayeva-Gupta et al., 2011). Although significant attention has historically focused on BRAF-mutant melanoma, NRAS-mutant melanoma represents a biologically distinct and therapeutically challenging subgroup (Johnson & Puzanov, 2015; Wellbrock & Arozarena, 2016). It has been observed that NRAS mutations tend to remain predominantly in their active GTP-bound conformation, mainly because codon 61 mutations impair the intrinsic enzymatic activity of NRAS to hydrolyze GTP to GDP (Li & Zhang, 2004; Vetter & Wittinghofer, 2001). The P-loop (also called the Walker-A or G1 motif) is a glycine-rich phosphate-binding loop that interacts with the β - and γ -phosphates of GDP/GTP and is highly conserved in Ras and other small GTPases (Vetter & Wittinghofer, 2001). This makes the task of drugging NRAS particularly difficult. Although mutation-specific targeting strategies have recently achieved success in KRAS(G12C), analogous approaches for NRAS are less developed, particularly because codon 61 mutations do not introduce reactive cysteine residues suitable for covalent inhibition (Moore et al., 2020; Ostrem & Shokat, 2016). For instance, some inhibitors such as the potential inhibitor HM-387 have been recently designed in silico to target the NRAS-Q61 positively charged mutant R61, inducing the GTP-bound NRAS-Q61 oncogenic mutant toward an “off-like” state. Consequently, most current therapeutic strategies for NRAS-mutant melanoma rely on indirect targeting of downstream signaling pathways (Johnson & Puzanov, 2015; Wellbrock & Arozarena, 2016). MEK inhibitors such as binimetinib and trametinib have demonstrated partial clinical activity; however, response rates remain modest and resistance frequently develops (Johnson & Puzanov, 2015). Farnesyltransferase inhibitors and combination therapies have also been explored, but durable and mutation-specific inhibition of NRAS remains unresolved (Moore et al., 2020). Therefore, the development of novel approaches capable of directly or allosterically modulating mutant NRAS represents a significant unmet medical need. Direct pharmacological targeting of NRAS remains challenging due to its high affinity for GTP and the absence of deep, well-defined binding pockets on its surface, features that have historically led to its classification as an “undruggable” oncogene (Cox et al., 2014; McCormick, 2015; Moore et al., 2020). In this context, natural products represent an attractive and relatively underexplored source of bioactive chemical scaffolds. Owing to their structural diversity, stereochemical complexity, and evolutionary optimization for biological activity, natural compounds have historically contributed substantially to anticancer drug development (Harvey et al., 2015; Newman & Cragg, 2020). Importantly, many clinically approved anticancer agents are derived from natural products or their analogues, underscoring their value in oncology drug discovery (Newman & Cragg, 2020).

Computational structure-based approaches offer a powerful framework for exploring such possibilities. Molecular docking and virtual screening enable the systematic evaluation of ligand binding modes, interaction networks, and conformational stability at atomic resolution (Kitchen et al., 2004; Lionta et al., 2014). These techniques are particularly valuable for targets like NRAS, where experimental structural characterization of ligand-bound complexes remains limited. In the present study, we employed a structure-based computational strategy to identify natural compounds from the ZINC database with potential inhibitory activity against clinically relevant mutant NRAS variants (G12, G13, and Q61). Mutant protein models were generated and structurally validated, followed by large-scale virtual screening of natural compounds and molecular docking. Candidate compounds were evaluated based on predicted binding free energy and detailed interaction analysis to identify top-ranking molecules exhibiting stable hydrogen-bond and hydrophobic interactions with key functional residues in the NRAS active site.

2.1. Methods

2.1.1. Target identification

The KEGG database (Kanehisa & Goto, 2000) was used to study the melanoma pathway. Dysregulation of the MAPK pathway is very common in many cancers due to mutations in the RAS gene. The MAPK pathway is one of the major pathways involved in the progress of melanoma and so it was chosen for this study. It harbours BRAF, NRAS, ERK, and MEK as its components. About 28% of the mutations are seen in NRAS of this pathway (Cancer.Net Editorial Board, 2025).

2.1.2. Protein Preparation

The wild type NRAS structure with the best resolution and well curated in UniProt was selected. The PDB ID of this structure was 5UHV and UniProtKB ID was P01111. NRAS is a small GTPase structure. A clear difference between isoforms of NRAS is the binding of Ras Binding Domain (Johnson et al., 2017). NRAS has only the A chain associated with it. Hence, docking was carried out on the A chain of NRAS on the active site. The G-domain, catalysing GTP hydrolysis and mediating downstream signalling, is 95% conserved between the Ras isoforms (Johnson et al., 2017). The wild type NRAS has a total of 189 amino acids. The overall structure can be divided into a catalytic G domain, which is comprised of residues 1-86 and 87-166 (Johnson et al., 2017).

2.1.3. Protein Structure Validation

ProSA (Wiederstein & Sippl, 2007) is frequently used for protein structure validation. ProSA calculates an overall

quality score for a specific input structure. The z-score indicates overall model quality (Wiederstein & Sippl, 2007). Its value is displayed in a plot showing the z-scores of all protein chains in the current PDB. ProSA-web visualizes the 3D structure of the input protein using the molecule viewer Jmol. ProSA was used for validation of the structure of mutant NRAS built in SPDBV. PROCHECK (Laskowski et al., 1993), a model evaluation and validation tool, checks for the stereochemical quality of a protein structure producing residue-by-residue geometry plots. It includes PROCHECK-NMR for checking the quality of structures solved by NMR. The G-factor provides a measure of how normal or unusual the property is. It gives the Ramachandran plot to understand if the residues are in allowed or disallowed regions. PROCHECK was used for validation of the structure of mutant NRAS built in SPDBV (Laskowski et al., 1993).

2.1.4. Compound Library Curation and Filtering

The database used for obtaining natural compounds for screening was the ZINC Database (ZINC Database, 2025). Natural products occupy an important part of the small molecule space because they are recognized by at least two proteins: the end of their biosynthetic pathway and their evolutionary biological target. The natural compounds screened were obtained from the ZINC natural product library. FAF-Drugs4 (Lagorce et al., 2017) is a program for filtering large compound libraries prior to in silico screening or related modelling studies. The tool can perform computational prediction of some ADME-Tox properties (adsorption, distribution, metabolism, excretion, and toxicity) to assist hit selection before chemical synthesis or ordering (Lagorce et al., 2017). This tool was used to filter out compounds in the ZINC database to remove interfering compounds and retain lead compounds with comparatively small log P for further optimization. PAINS moieties, or Pan Assay Interference Compounds, are compounds that occur frequently in biochemical high throughput screens.

2.1.5. Virtual Screening and Molecular Docking

PyRx (Radhakrishnan et al., 2024) is a virtual screening tool used for docking natural compounds. It uses the AutoDock 4.2 suite with Lamarckian genetic algorithm. The tool was first evaluated using the NRAS mutant structure and the ligand phosphoaminophosphonic acid-guanylate ester, a non-hydrolyzable analog of GTP. The nucleotide is a potent stimulator of adenylate cyclase and was docked. The RMSD tolerance of 2.0 Å indicated good software stability. The grid box was set to 50 × 50 × 50 with a spacing of 0.3750 Å. All ligands were converted to pdbqt format in PyRx. AutoDock was set to run generating 10 poses per compound. Only the pose with the best binding score was tabulated in the sdf files. This yielded the best protein-ligand binding candidates

for further analysis. Docking results were analysed using LigPlot+ (LigPlus, 2025), an advanced version of LIGPLOT for automatic generation of 2D ligand-protein interaction diagrams. The hydrophobic and hydrogen bond interactions contribute to protein-ligand complex stability and help determine the inhibitory potential of each compound on mutant NRAS.

3. RESULTS and DISCUSSION

3.1. Validation of mutant NRAS models

The mutant NRAS variants (G12D, G13D, and Q61R) were generated using SwissPDB Viewer based on the wild-type structural template. Structural quality checks confirmed the reliability of these models. The protein model's backbone torsion angles (phi and psi) are highly stereochemically favourable, over 90% of residues falling in the most allowed regions of the Ramachandran plot (Figure 1E–F), signifying excellent structural quality. Only a minor fraction occupied disallowed regions. ProSA scores were negative, indicating minimal erroneous regions (Figure 1C) were consistent with native proteins of similar size, indicating reliable global folding (Wiederstein & Sippl, 2007). The modelled NRAS mutant structure was therefore used for further analysis, demonstrating strong structural stability.

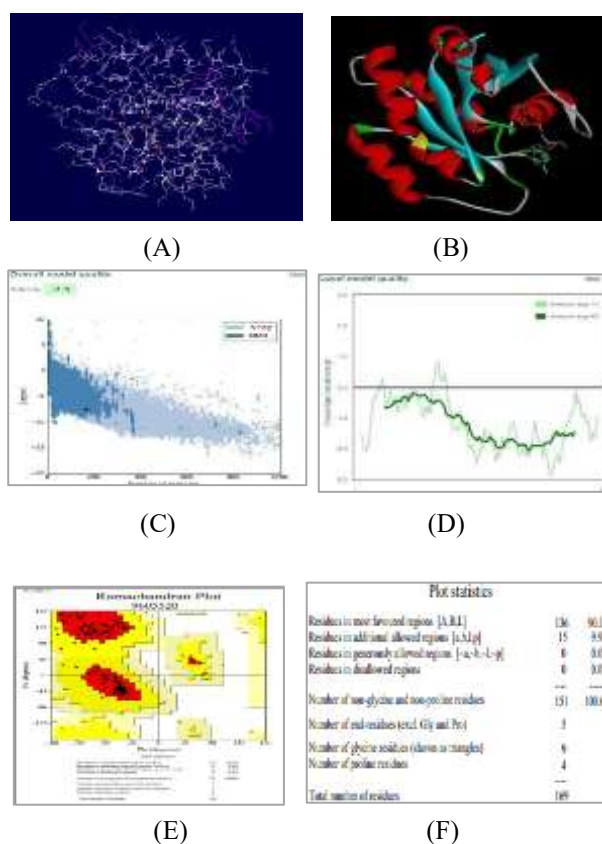


Figure 1. (A) Protein showing secondary structure elements and residue connectivity. (B) Cartoon

representation of the protein and colored by secondary structure type. (C) Overall model quality plot in PROSA indicating global Z-scores (D) Local quality assessment plot showing per-residue score fluctuations (E) Ramachandran plot presenting backbone dihedral angle distributions, with most residues occupying favored regions. (F) Summary of model statistics obtained from Ramachandran Plot.

3.2. Virtual screening and docking outcomes

Structure based virtual screening is a widely accepted early stage drug discovery strategy for identifying lead candidates against challenging targets (Kitchen et al., 2004; Lionta et al., 2014). We conducted a campaign of 5,322 natural compounds from the ZINC database, filtered through FAFDrugs4 to exclude compounds violating Lipinski's rules (Lagorce et al., 2017). Docking was performed with AutoDock 4.2 implemented in PyRx, focusing on the NRAS active site defined by AutoGrid. Following screening, eleven top-ranked compounds were selected based on their binding free energy, with docking scores ranging from -10.62 to -10.04 kcal/mol (Table 1). These values compare favorably with reported docking energies for small-molecule interactions targeting RAS proteins (Cox et al., 2014), suggesting stable and energetically favorable complex formation. The narrow energy distribution among the top hits indicates convergence toward a structurally consistent binding mode within the NRAS active site. Importantly, the identification of multiple natural compounds with strong predicted affinity inclines to the therapeutic potential of natural products chemical space for targeting oncogenic NRAS variants.

3.3. Conserved residue interactions

Each panel in Figure 2 presents a detailed schematic of the non-covalent binding environment for a specific

ligand within its target protein structure, with both the ligand and surrounding residues depicted in ball-and-stick format for clarity at atomic resolution shown in. Hydrogen bonding interactions are marked by dashed green lines linking donor and acceptor atoms, while hydrophobic contacts are highlighted by spoked red arcs oriented toward the ligand, illustrating side-chain proximity and surface complementarity. This standardized visualization enables straightforward comparison of key contacts including hydrogen bonds and hydrophobic interactions among different ligand binding poses and mutant protein variants, providing insight into structural determinants of binding affinity. Interaction analysis of the docked complexes revealed frequently conserved functional residues that are significant to NRAS catalytic activity and conformational stability. Specific residues like Tyr-32, located within the Switch I region contributes hydrophobic stabilization in five complexes (compounds 1, 3, 7, 8, and 9). Switch I undergoes nucleotide-dependent conformational rearrangement and plays a critical role in effector identification and Mg²⁺ coordination while when we look at Thr-35, another Switch I residue, forms a stable hydrogen bond with compound 2 and engages itself in hydrophobic contacts in complexes 9 and 11. Phe-28, positioned adjacent to the P-loop and contributing to the structural integrity of the nucleotide-binding pocket, participated in hydrophobic interactions in seven complexes. This high frequency emphasizes its structural role in maintaining the architecture of the protein and suggests that ligand binding consistently overlaps with the nucleotide-binding site. Similarly, Asp-119, a residue involved in nucleotide specificity and stabilization, shows consistent engagement in nine complexes, further indicating that the compounds target catalytically relevant regions.

Table 1. "Docking results of top-ranking ZINC compounds (binding free energy ≤ -10 kcal/mol), highlighting predicted hydrogen bond donor-acceptor pairs and residues involved in hydrophobic or van der Waals interactions with the protein active site."

Sr. No.	ZINC ID	Binding free Energy (kcal/mol)	Hydrogen bonding pairs HD::HA	Residues in Hydrophobic Interactions or van der Waals contact
1	ZINC02107288	-10.62	Lys-16(N):: Unk0(C)OXT Lys-16(NZ):: Unk0(C)OXT Gly-15(N):: Unk0(C)OXT Ser-17(N)::Unk0(C)O Asp-33(N)::Unk0(C11)O	Val-14, Asp-13, Gly-60, Asp-12, Tyr-32, Glu-31, Ala-18, Asn-116, Lys-117, Asp-119, Lys-147, Phe-28, Pro-34
2	ZINC02096813	-10.61	Thr-35(N):: Unl1(C23) O5 Asp-12(OD1):: Unl1(N) Gly-60(N):: Unl1(OXT) Lys-16(NZ):: Unl1(OXT)	Ser-17, Gly-15, Asn-116, Lys-117, Phe-28, Lys-147, Ala-146, Asp-119, Ser-145, Ala-18, Tyr-32, Asp-13
3	ZINC02096815	-10.33	Lys-16(N):: Unl1(C23)O5 Lys-16(NZ):: Unl1(C23)OXT Gly15(N):: Unl1(C23)O5	Asp-12, Thr-58, Asp-13, Asp-33, Try-32, Glu-31, Phe-28, Lys-147, Asp-119, Ala-146,

			Ser17(OG):: Unl1(C20)O4	Lys-117, Asn-116, Thr-35
4	ZINC02114326	-10.23	Asn-116(ND2):: Unl1(CL) Lys-147(NZ):: Unl1(OXT) Asp-33(N):: Unl1 (C12) O4	Phe-28, Asp-119, Leu-120, Gly-15, Lys-117, Asp-30, Lys-16, Ser-17, Ala-18, Tyr-32, Glu-31
5	ZINC01020381	-10.16	Gly-15(N):: Unl1(C24) O2 Lys-16(N):: Unl1(C24) O2 Lys-16(NZ):: Unl1(C24) O2	Asp-119, Lys-147, Asn-116, Ala-146, Ser-145, Glu-31, Phe-28, Asp-30, Asp-13, Thr-58, Asp-33
6	ZINC06167356	-10.16	Thr-58(O)::Unl1(C6) O6 Gly-60(N):: Unl1(C6) O6 Lys-117(NZ):: Unl1(C18) O5 Asp-30(N):: Unl1(C16) O4 Glu-31(N):: Unl1(C16) O 4 Ala-18(N):: Unl1(C9) O 2 Ser-17(N):: Unl1(O1)	Asp-33, Asp-12, Lys-16, Thr-35, Gly-15, Pro-34, Ala-59, Asp-13, Val-29
7	ZINC02120343	-10.14	Asp-33(N):: Unl1(C9) O2 Ser-17(N):: Unl1(C22) O5 Lys-16(N):: Unl1(C22) O4 Lys-16(NZ):: Unl1(C22) O 4 Asp-13(N):: Unl1(C22) O4 Gly-15(N):: Unl1(C22) O4	Asp-119, Ser-145, Lys-147, Ala-146, Phe-28, Ala-18, Asn-116, Lys-117, Glu-31, Tyr-32, Pro-34, Asp-12, Val-14
8	ZINC02102162	-10.14	Lys-16(N):: Unl1(C22) O6 Lys-16(NZ):: Unl1(C22) O5 Val-14(N):: Unl1(C22) O5 Gly-15(N):: Unl1(C22) O5 Ser-17(N):: Unl1(C22) O6	Pro-34, Asp-12, Asp-13, Asp-33, Glu-31, Tyr-32, Phe-28, Lys-117, Asn-116, Ala-18, Ala-146, Lys-147, Asp-119
9	ZINC02114643	-10.1	Asp-33(N)::Unl1(C16) O4 Lys-16(N)::Unl1(C24) O5 Lys-16(NZ)::Unl1(C24) O6 Gly-15(N)::Unl1(C24) O5	Asp-19, Phe-28, Lys-147, Lys-147, Leu-120, Tyr-32, Ala-18, Ala-146, Lys-117, Thr-35, Glu-31, Pro-34, Asp-13
10	ZINC00518885	-10.06	Lys-117(N):: Unk0(C9) O2 Ala-146(N):: Unk0(C9) O2 Asn-116(ND2)::Unk0(O3) Asp-13(N):: Unk0(OXT) Lys-16(N)::Unk0(O) Lys-16(NZ)::Unk0(O) Gly-15(N)::Unk0(O) Val-14(N)::Unk0(O)	Asp-119, Phe-28, Asp-33, Pro-34, Asp-12, Ala-18, Ser-17, Lys-147, Ser-145
11	ZINC03882094	-10.04	Lys-16(NZ)::Unl1(C21) O3 Gly-15(N):: Unl1(C21) O3 Ser-17(N)::Unl1(C22) O4	Thr-35, Asp-13, Val-14, Asp-12, Asp-30, Glu-13, Asp-119, Lys-117, Phe-28, Asp-57, Ala-18, Thr-58

3.3.1 Mutant Residue Contributions

Beyond conserved sites, mutant residues also established direct ligand interactions. The variants, Asp-12 (G12D) formed a hydrogen bond in compound 2 (2.88 Å) and engaged in additional stabilizing contacts across multiple complexes. Asp-13 (G13D) formed hydrogen bonds in compounds 7 (3.05 Å) and 10 (2.84 Å), while Arg-61 (Q61R), located in Switch II region, not that common, participated in electrostatic and hydrogen-bond interactions depending on ligand orientation.

3.3.2 Lead Compound Identification

Among the eleven shortlisted compounds, ZINC02096813, ZINC02120343, and ZINC00518885 consistently demonstrated strong binding affinities. These compounds simultaneously interact with catalytic stabilisers (Tyr-32, Phe-28, Asp-119) and mutant residues (Asp-12, Asp-13, or Arg-61), forming complementary networks of hydrogen bonds and hydrophobic contacts, this anchoring likely enhances binding stability. This dual engagement of hydrogen bonds with mutant residues alongside hydrophobic contacts with conserved structural stabilisers suggests enhanced stability and specificity of binding. These compounds therefore represent the most promising leads

for further optimization and in vitro testing against NRAS-driven melanoma. Collectively, these observations suggest that gain-of-function mutations in NRAS do more than alter intrinsic catalytic properties, they actively reshape the protein's binding pocket, creating new places for ligand engagement that are absent in the wild-type context. **This synergy of interactions**

enhances both the stability and specificity of binding, suggesting potential for mutation-biased binding that warrants further experimental validation. Taken together, these results underscore the value of targeting both conserved and mutant-specific interactions for inhibitor design. Their unique binding profiles suggest they may exploit structural features distinctive to mutant NRAS.

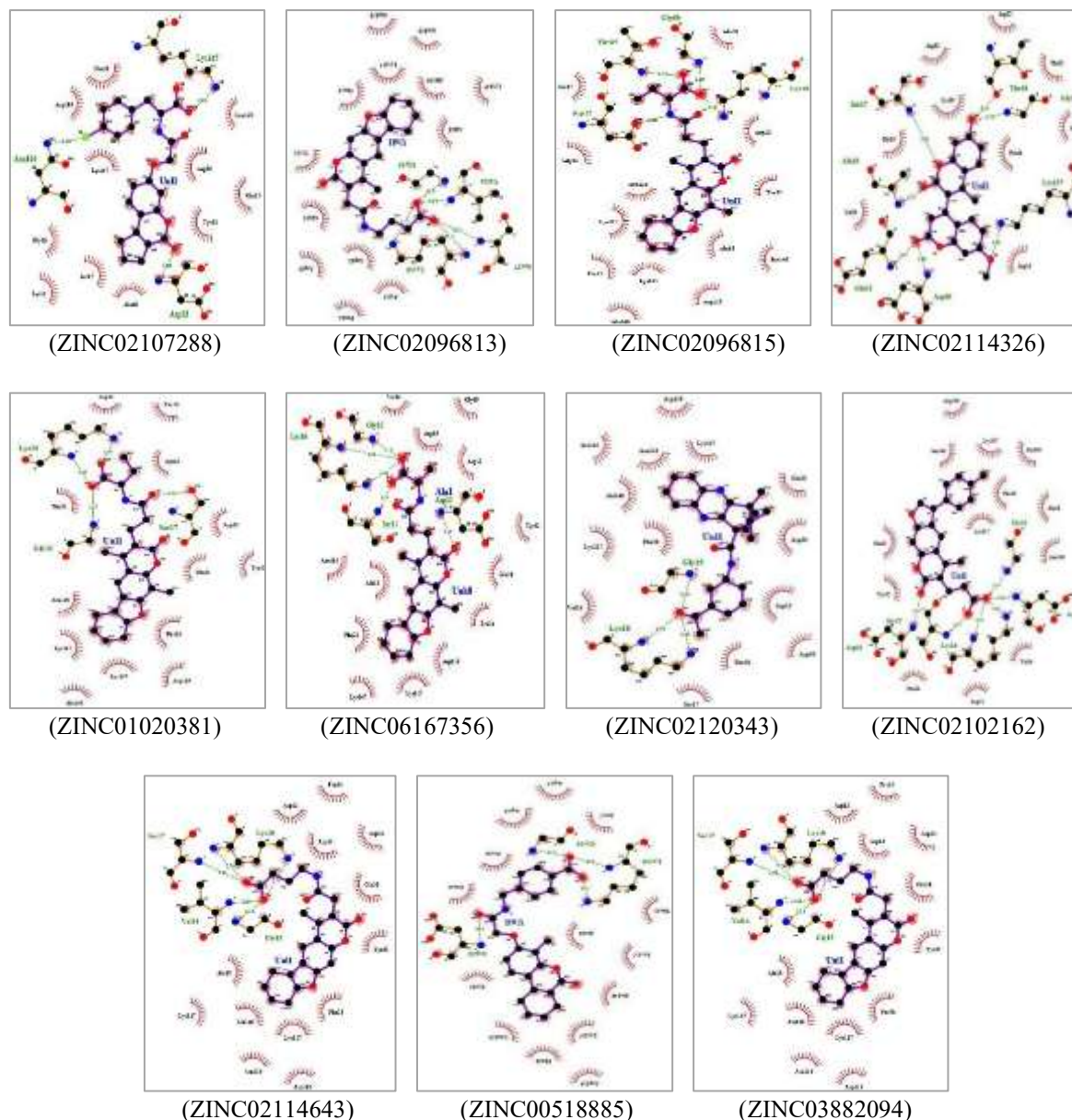


Figure 2. Two-dimensional interaction diagrams of the top eleven NRAS ligand complexes generated using LigPlot+. Hydrogen bonds are represented as dashed green lines, while hydrophobic contacts are illustrated as red semicircular arcs. The diagrams highlight conserved and mutant residue interactions within the nucleotide-binding pocket, demonstrating consistent engagement of P-loop and switch-region residues across the highest-ranked ZINC compounds.

Such mutation-dependent contacts suggest local remodeling of the binding pocket, potentially generating novel interaction opportunities. This phenomenon parallels findings in KRAS G12C targeting, where

mutation specific pockets enabled selective inhibitor development [Ostrem et al., 2013]. The dual engagement of conserved catalytic stabilizers and mutation-specific

residues suggests a rational framework for designing selective NRAS inhibitors.

4. DISCUSSION

NRAS being involved in one of the most important signaling pathway called MAPK (Wellbrock & Arozarena, 2016) and despite it's clinical importance, direct pharmacological inhibition remains limited and that are yet to be explored. Current therapeutic strategies for NRAS mutant melanoma primarily rely on downstream MEK inhibition (Johnson & Puzanov, 2015). However, clinical responses to MEK inhibitors such as binimetinib have demonstrated modest improvements in progression free survival (Ascierto et al., 2013). The structural challenges seen in RAS family proteins have historically been considered "undruggable" due to their high affinity for GTP/GDP and the absence of deep hydrophobic pockets (Cox et al., 2014; McCormick, 2015). Direct NRAS targeting, as explored in this study, may therefore complement or enhance these approaches by addressing the root cause. Natural products account for a substantial proportion of approved anticancer drugs (Newman & Cragg, 2020). Their structural diversity and evolutionary optimization enable interactions with challenging protein surfaces (Harvey et al., 2015). Our results suggest that natural compounds, with diverse scaffolds and biocompatibility, may offer new chemical starting points to a strategic target in finding novel inhibitors for mutant NRAS. The diversity observed among the top-ranked compounds in this study further underscores the chemical richness of natural product libraries and their suitability for challenging oncogenic targets such as NRAS. Our analysis highlights the importance of conserved residues, which are known to stabilize switch I/II regions and coordinate nucleotide binding. The observation that these residues consistently contributed to ligand stabilization agrees with prior crystal and biochemical studies on RAS proteins.

Although NRAS lacks the reactive cysteine exploited in KRAS G12C inhibitors, these breakthroughs provide conceptual validation that mutation-specific pockets can be therapeutically elevated. In this context, our findings support the possibility that NRAS G12 and G13 substitutions may create microenvironments capable of accommodating small molecules, particularly those capable of forming stable hydrogen-bonding networks. Furthermore, accumulating evidence suggests that allosteric modulation of RAS proteins may represent a viable alternative to direct nucleotide competitive inhibition (Maurer et al., 2012; Welsch et al., 2017). The interaction patterns observed in our analysis, particularly near the switch I and switch II regions, are consistent with previously described transient pockets that regulate effector binding. Nonetheless, the findings presented here are limited by their computational nature. Docking captures a static snapshot of protein–ligand interactions and does not fully address conformational flexibility.

(Kitchen et al., 2004). Future studies incorporating molecular dynamics simulations, free energy calculations would strengthen our understanding of the structural details and energetics of the system that can further enhance the binding predictions (Ferreira et al., 2015). Later Experimental validation through biochemical assays and melanoma cell models will be necessary to validate our predictions. Collectively, this study contributes to the expanding efforts aimed at overcoming the challenges associated with direct NRAS inhibition, by integrating natural product diversity with structure-based virtual screening.

4.CONCLUSION

This study integrates structural modelling, virtual screening and docking to identify potential inhibitors of oncogenic NRAS variants. The mutant models built exhibited high stereochemical quality, enabling reliable analyses. Screening 5,322 natural compounds yielded three guided candidates namely (ZINC02096813, ZINC02120343, and ZINC00518885). They followed strong binding affinities and dual involvement of conserved catalytic residues (Tyr-32, Phe-28, Asp-119) along with mutant sites (Asp-2, Asp-13, Arg-61). These findings suggest that mutation-induced structural plasticity within NRAS can be strategically exploited for inhibitor development. These results being the initial steps towards finding novel inhibitor, they provide a foundation for further molecular dynamics simulations and experimental validation towards therapies for NRAS driven melanoma.

Acknowledgements

The authors gratefully acknowledge Dr. Rohan Meshram from Savitribai Phule Pune University for his invaluable guidance and support throughout the study.

Conflict of Interest

The authors declare that they have no conflicts of interest.

Table 1. "Docking results of top-ranking ZINC compounds (binding free energy ≤ -10 kcal/mol), highlighting predicted hydrogen bond donor–acceptor pairs and residues involved in hydrophobic or van der Waals interactions with the protein active site."

Figure 1. (A) Protein showing secondary structure elements and residue connectivity. (B) Cartoon representation of the protein and colored by secondary structure type. (C) Overall model quality plot in PROSA indicating global Z-scores (D) Local quality assessment plot showing per-residue score fluctuations (E) Ramachandran plot presenting backbone dihedral angle distributions, with most residues occupying favored

regions. (F) Summary of model statistics obtained from Ramachandran Plot.

Figure 2. Two-dimensional interaction diagrams of the top eleven NRAS ligand complexes generated using LigPlot+. Hydrogen bonds are represented as dashed green lines, while hydrophobic contacts are illustrated as red semicircular arcs. The diagrams highlight conserved and mutant residue interactions within the nucleotide-binding pocket, demonstrating consistent engagement of P-loop and switch-region residues across the highest-ranked ZINC compounds.

REFERENCES

- Al Mahi, A., & Ablain, J. (2022). RAS pathway regulation in melanoma. *Disease Models & Mechanisms*, 15(2), dmm049229. <https://doi.org/10.1242/dmm.049229>
- Ardito, C. M., Grüner, B. M., Takeuchi, K. K., Lubeseder-Martellato, C., Teichmann, N., Mazur, P. K., DelGiorno, K. E., Carpenter, E. S., Halbrook, C. J., Hall, J. C., et al. (2012). EGF receptor is required for KRAS-induced pancreatic tumorigenesis. *Cancer Cell*, 22(3), 304–317. <https://doi.org/10.1016/j.ccr.2012.07.024>
- Ascierto, P. A., Schadendorf, D., Berking, C., Agarwala, S. S., van Herpen, C. M., Queirolo, P., Blank, C. U., Hauschild, A., Beck, J. T., St-Pierre, A., et al. (2013). MEK162 for patients with advanced melanoma harbouring NRAS or BRAF mutations: A non-randomised, open-label phase 2 study. *The Lancet Oncology*, 14(3), 249–256. [https://doi.org/10.1016/S1470-2045\(13\)70024-X](https://doi.org/10.1016/S1470-2045(13)70024-X)
- Cancer.Net Editorial Board. (2025a). Melanoma: Stages. American Society of Clinical Oncology.
- Cancer.Net Editorial Board. (2025b). Melanoma: Statistics. American Society of Clinical Oncology.
- Cherfils, J., & Zeghouf, M. (2013). Regulation of small GTPases by GEFs, GAPs, and GDIs. *Physiological Reviews*, 93(1), 269–309. <https://doi.org/10.1152/physrev.00003.2012>
- Cox, A. D., Fesik, S. W., Kimmelman, A. C., Luo, J., & Der, C. J. (2014). Drugging the undruggable RAS: Mission possible? *Nature Reviews Drug Discovery*, 13(11), 828–851. <https://doi.org/10.1038/nrd4389>
- Ferreira, L. G., Dos Santos, R. N., Oliva, G., & Andricopulo, A. D. (2015). Molecular docking and structure-based drug design strategies. *Molecules*, 20(7), 13384–13421. <https://doi.org/10.3390/molecules200713384>
- Harvey, A. L., Edrada-Ebel, R., & Quinn, R. J. (2015). The re-emergence of natural products for drug discovery in the genomics era. *Nature Reviews Drug Discovery*, 14, 111–129. <https://doi.org/10.1038/nrd4510>
- Howe, L. R., Leever, S. J., Gómez, N., Nakielnny, S., Cohen, P., & Marshall, C. J. (1992). Activation of the MAP kinase pathway by the protein kinase Raf. *Cell*, 71(2), 335–342. [https://doi.org/10.1016/0092-8674\(92\)90362-3](https://doi.org/10.1016/0092-8674(92)90362-3)
- Hu, Z., & Martí, J. (2025). Atomic-level mechanisms of abnormal activation in NRAS oncogenes from two-dimensional free energy landscapes. *Nanoscale*, 17. <https://doi.org/10.1039/D4NR03372H>
- Johnson, C. W., Reid, D., Parker, J. A., Harrison, R. A., Weston, L., Peti, W., et al. (2017). The small GTPases K-Ras, N-Ras, and H-Ras have distinct biochemical properties determined by allosteric effects. *Journal of Biological Chemistry*, 292(31), 12981–12993. <https://doi.org/10.1074/jbc.M117.778886>
- Johnson, D. B., & Puzanov, I. (2015). Treatment of NRAS-mutant melanoma. *Current Treatment Options in Oncology*, 16(4), 15. <https://doi.org/10.1007/s11864-015-0330-z>
- Kanehisa, M., & Goto, S. (2000). KEGG: Kyoto encyclopedia of genes and genomes. *Nucleic Acids Research*, 28(1), 27–30. <https://doi.org/10.1093/nar/28.1.27>
- Kitchen, D. B., Decornez, H., Furr, J. R., & Bajorath, J. (2004). Docking and scoring in virtual screening for drug discovery. *Nature Reviews Drug Discovery*, 3(11), 935–949. <https://doi.org/10.1038/nrd1549>
- Kolch, W., Berta, D., & Rosta, E. (2023). *Biochemical Journal*, 480, 1–23. <https://doi.org/10.1042/BCJ20220547>
- Lagorce, D., Bouslama, L., Becot, J., Ménard, P., & Villoutreix, B. O. (2017). FAF-Drugs4: Free ADME-tox filtering computations for chemical biology and early stages drug discovery. *Bioinformatics*, 33(22), 3658–3660. <https://doi.org/10.1093/bioinformatics/btx491>
- Li, G., & Zhang, X. C. (2004). GTP hydrolysis mechanism of Ras-like GTPases. *Journal of Molecular Biology*, 340(5), 921–932. <https://doi.org/10.1016/j.jmb.2004.05.079>
- Li, S., Balmain, A., & Counter, C. M. (2018). A model for RAS mutation patterns in cancers: Finding the sweet

spot. *Nature Reviews Cancer*, 18(12), 767–777.
<https://doi.org/10.1038/s41568-018-0076-6>

LigPlus. (2025). Automatic generation of ligand–protein interaction diagrams. European Bioinformatics Institute.

Lionta, E., Spyrou, G., Vassilatis, D. K., & Cournia, Z. (2014). Structure-based virtual screening for drug discovery: Principles, applications and recent advances. *Current Topics in Medicinal Chemistry*, 14(16), 1923–1938.
<https://doi.org/10.2174/1568026614666140929124445>

Maurer, T., Garrenton, L. S., Oh, A., Pitts, K., Anderson, D. J., Skelton, N. J., Fauber, B. P., Pan, B., Malek, S., Stokoe, D., et al. (2012). Small-molecule ligands bind to a distinct pocket in Ras and inhibit SOS-mediated nucleotide exchange activity. *Proceedings of the National Academy of Sciences USA*, 109(14), 5299–5304. <https://doi.org/10.1073/pnas.1116510109>

McCormick, F. (2015). KRAS as a therapeutic target. *Clinical Cancer Research*, 21, 1797–1801.
<https://doi.org/10.1158/1078-0432.CCR-14-2662>

Moore, A. R., Rosenberg, S. C., McCormick, F., & Malek, S. (2020). RAS-targeted therapies: Is the undruggable drugged? *Nature Reviews Drug Discovery*, 19(8), 533–552. <https://doi.org/10.1038/s41573-020-0068-6>

National Cancer Institute. (2025). The cancer genome atlas (TCGA) – Melanoma skin cancer.

Navas, C., Hernández-Porras, I., Schuhmacher, A. J., Sibilia, M., Guerra, C., & Barbacid, M. (2012). EGF receptor signaling is essential for K-Ras oncogene-driven pancreatic ductal adenocarcinoma. *Cancer Cell*, 22(3), 318–330.
<https://doi.org/10.1016/j.ccr.2012.07.024>

Newman, D. J., & Cragg, G. M. (2020). Natural products as sources of new drugs. *Journal of Natural Products*, 83(3), 770–803.
<https://doi.org/10.1021/acs.jnatprod.9b01285>

Normanno, N., Tejpar, S., Morgillo, F., De Luca, A., Van Cutsem, E., & Ciardiello, F. (2009). Implications for KRAS status and EGFR-targeted therapies in metastatic colorectal cancer. *Nature Reviews Clinical Oncology*, 6(9), 519–527.
<https://doi.org/10.1038/nrclinonc.2009.111>

Ostrem, J. M., & Shokat, K. M. (2016). Direct small-molecule inhibitors of KRAS: From structural insights to mechanism-based design. *Nature Reviews Drug Discovery*, 15(11), 771–785.
<https://doi.org/10.1038/nrd.2016.139>

PROCHECK. (2025). Stereochemical quality assessment of protein structures. European Bioinformatics Institute.

ProSA. (2025). Protein structure analysis.

Pylayeva-Gupta, Y., Grabocka, E., & Bar-Sagi, D. (2011). RAS oncogenes: Weaving a tumorigenic web. *Nature Reviews Cancer*, 11(11), 761–774.
<https://doi.org/10.1038/nrc3106>

Radhakrishnan, S., Menon, P., & Silva, K. (2024). Python Prescription (PyRx) transforms virtual drug discovery with AI-driven tools. *African Journal of Biomedical Research*, 27(3s).

Rodriguez-Viciano, P., Warne, P. H., Dhand, R., Vanhaesebroeck, B., Gout, I., Fry, M. J., Waterfield, M. D., & Downward, J. (1994). Phosphatidylinositol-3-OH kinase as a direct target of Ras. *Nature*, 370(6490), 527–532. <https://doi.org/10.1038/370527a0>

Spoerner, M., Hozsa, C., Poetzel, J. A., Reiss, K., Ganser, P., Geyer, M., & Kalbitzer, H. R. (2010). Conformational states of human Ras protein complexed with GTP. *Journal of Biological Chemistry*, 285(51), 39768–39778. <https://doi.org/10.1074/jbc.M110.145235>

Vetter, I. R., & Wittinghofer, A. (2001). The guanine nucleotide-binding switch in three dimensions. *Science*, 294(5545), 1299–1304.
<https://doi.org/10.1126/science.1062023>

Wellbrock, C., & Arozarena, I. (2016). The complexity of the ERK/MAP-kinase pathway and the treatment of melanoma skin cancer. *Frontiers in Cell and Developmental Biology*, 4, 33.
<https://doi.org/10.3389/fcell.2016.00033>

Welsch, M. E., Kaplan, A., Chambers, J. M., Stokes, M. E., Bos, P. H., Zask, A., Zhang, Y., Sanchez-Martin, M., Badgley, M. A., Huang, C. S., et al. (2017). Multivalent small-molecule pan-RAS inhibitors. *Cell*, 168(5), 878–889. <https://doi.org/10.1016/j.cell.2017.02.006>

Wood, K. W., Sarnecki, C., Roberts, T. M., & Blenis, J. (1992). Ras mediates nerve growth factor receptor modulation of three signal-transducing kinases. *Cell*, 68(6), 1041–1050. [https://doi.org/10.1016/0092-8674\(92\)90072-8](https://doi.org/10.1016/0092-8674(92)90072-8)

ZINC Database. (2025). Ligand discovery and virtual screening.